

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks Available |     |     |       |       |      |
|----------|-----|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 4        | (a) |  |  | <p><b>Indicative content:</b></p> <ol style="list-style-type: none"><li>1. Set the height of the ramp to about 10cm (above the desk)</li><li>2. Measure a distance of 50cm from the end of the ramp and mark this point.</li><li>3. Release the squash ball from the top of the ramp starting the stopwatch as you do.</li><li>4. When the squash ball reaches the bottom of the ramp press the lap button on the stopwatch.</li><li>5. Stop the stopwatch when the squash ball reaches the 50cm mark. Candidate may say there are 2 people using 2 stopwatches.</li><li>6. Record the time taken for the ball to travel down the ramp (lap time) and the time to travel 50 cm.</li><li>7. To reduce uncertainty could:<ul style="list-style-type: none"><li>• repeat experiment (twice more)/compare with different groups</li><li>• use longer ramp</li><li>• increase the 50 cm distance</li><li>• reduce angle of ramp</li><li>• roll from same point on ramp</li><li>• same person timing</li><li>• record the experiment/light gate</li></ul></li></ol> <p><b>5-6 marks</b><br/>Detailed description including both time measurements, sensible distances and how to reduce uncertainty.<br/><i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i></p> | 6               |     |     | 6     |       | 6    |